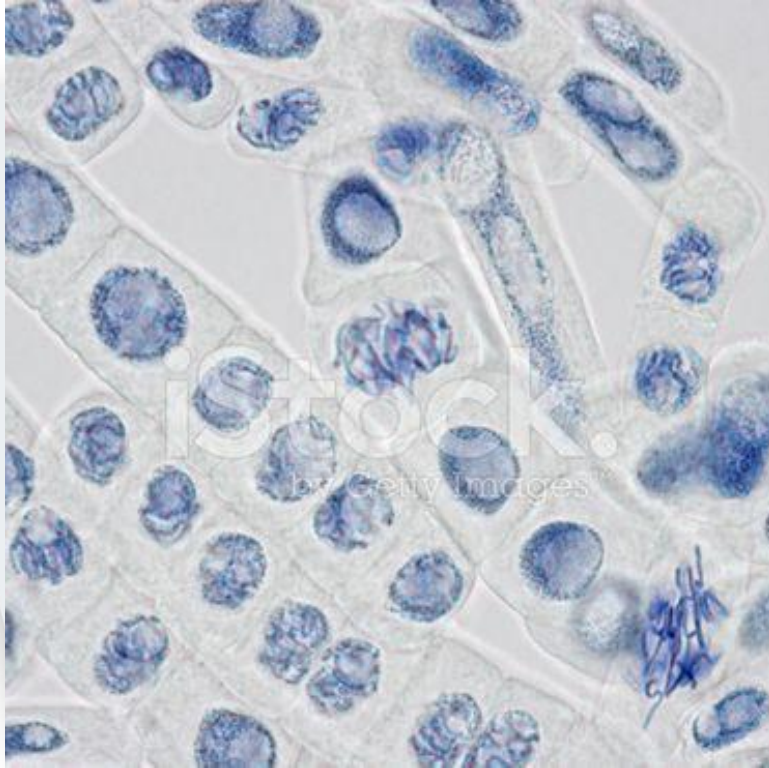




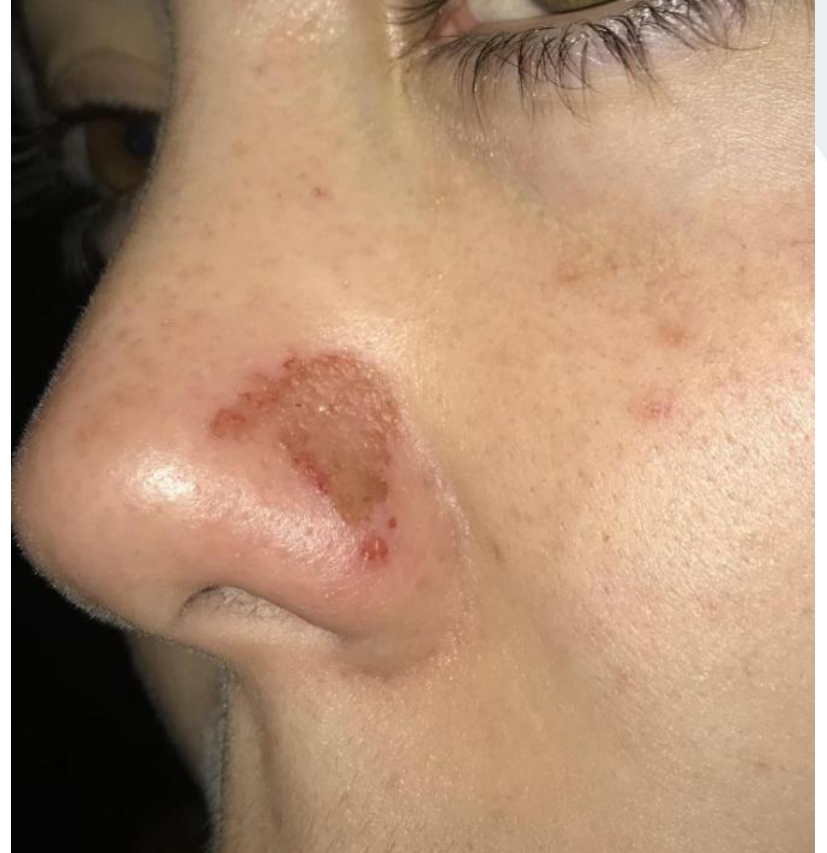
Mitosis: The Cycle of Life

Cell division and the growth of organisms.

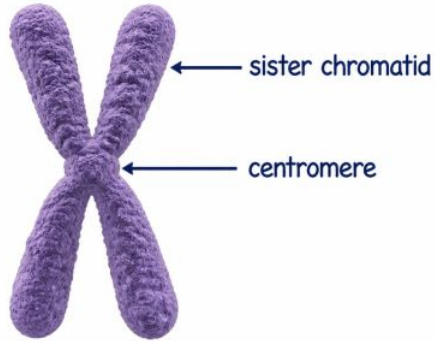
Learning Objectives

By the end of this lesson, you will be able to:

- Describe the stages of the **cell cycle**.
- Explain why **mitosis** is essential for growth and repair.
- Identify the key features of each phase of mitosis.
- Understand the production of **genetically identical** daughter cells.

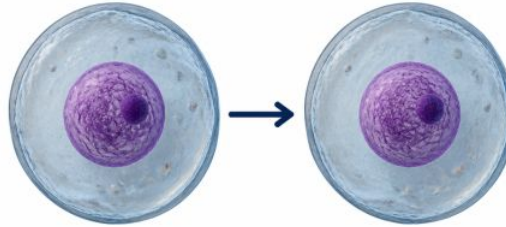


Key Vocabulary



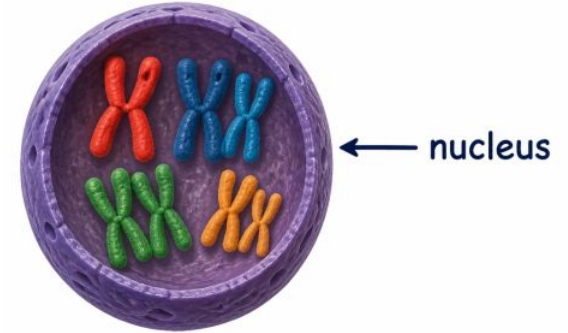
Chromosome

A thread-like structure of DNA and protein found in the nucleus.



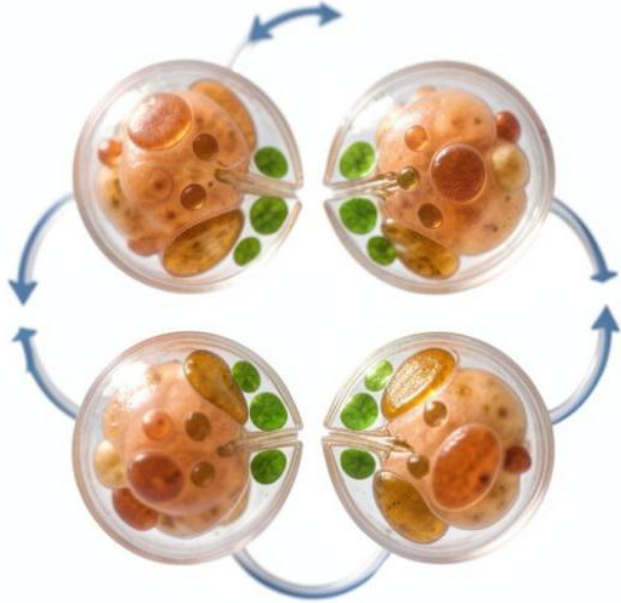
Daughter Cell

One of the two cells produced when a parent cell divides.



Diploid

A cell containing two complete sets of chromosomes, one from each parent.



How did you get here?

Every human begins as a single fertilised egg cell, known as a **zygote**.

How do we go from one single cell to the trillions of cells that make up your body today? The answer lies in the precision of the cell cycle.

Key point

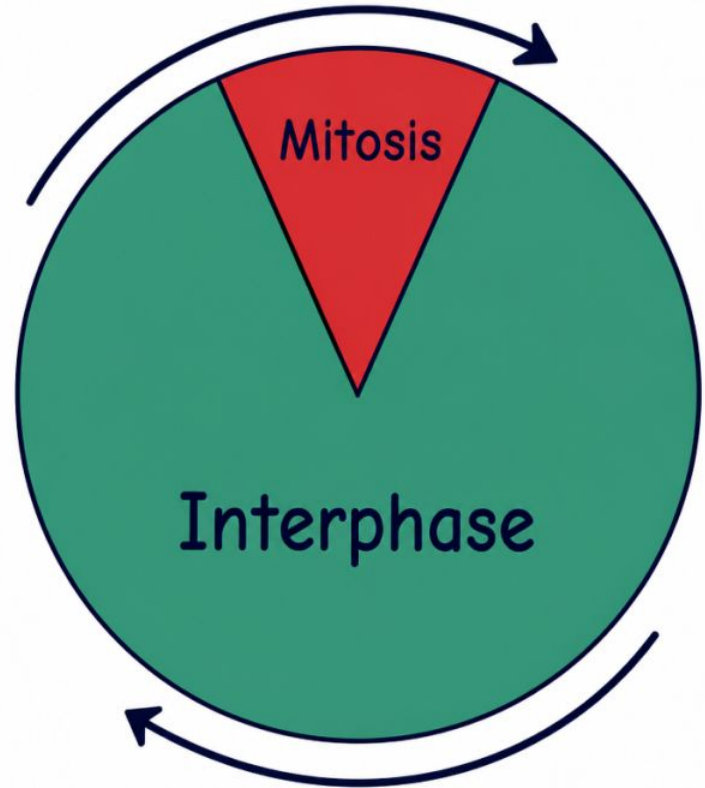
Mitosis doesn't just make more cells; it makes *exact copies*.

The Cell Cycle

KEY CONCEPT

Before a cell can divide, it must grow and prepare. This is called **Interphase**.

1. The cell grows in size.
2. It increases the number of sub-cellular structures like mitochondria and ribosomes.
3. It **replicates its DNA** to form two copies of each chromosome.



Check Your Understanding

Answers on the next slide...

What must happen to the DNA before a cell can divide?

1. It must be destroyed.
2. It must be replicated (copied).
3. It must be halved.
4. It stays exactly the same without changing.

Check Your Understanding



What must happen to the DNA before a cell can divide?

1.

It must be destroyed.

2.

It must be replicated (copied).

3.

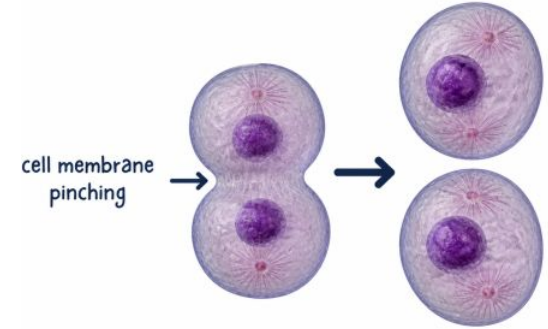
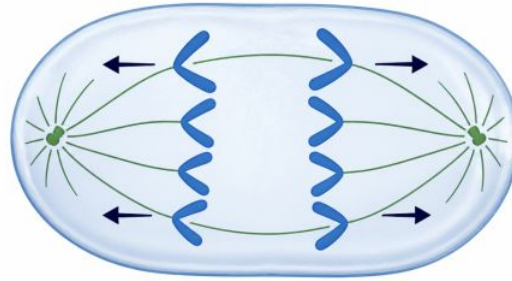
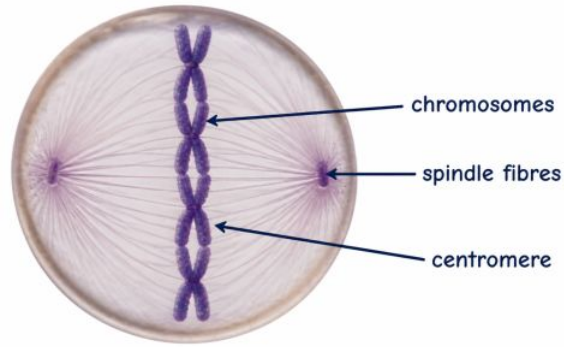
It must be halved.

4.

It stays exactly the same without changing.

The Stages of Mitosis

Mitosis occurs in four stages: Prophase, Metaphase, Anaphase, and Telophase, ending with cell division.



Prophase & Metaphase:
Chromosomes condense and align at the cell center.

Anaphase: Spindle fibres pull chromosome arms to opposite cell poles.

Telophase & Cytokinesis:
Nuclei reform and the cell membrane pinches to divide.

Mitosis in Motion

Watch this animation to see how the chromosomes are carefully sorted to ensure each new cell gets a full set of instructions.



Why is Mitosis Critical?



DISCUSSION

Mitosis occurs for three main reasons:

- **Growth:** Increasing the number of cells in multicellular organisms.
- **Repair:** Replacing damaged or dead cells (like skin or blood cells).
- **Asexual Reproduction:** Some organisms, like strawberries or yeast, reproduce this way.

Example

Red blood cells only live for 120 days. Your body uses mitosis to create millions of new ones every second!

The Root Tip Experiment

Listen to the audio and decide: true or false?



Laboratory Discussion on Cell Division

▶ Play audio

1 👍 | 👎 Onion root tips are used because they have a high rate of cell division.

2 👍 | 👎 Most cells under the microscope will be seen in the middle of dividing.

3 👍 | 👎 Sarah identifies a cell with chromosomes lined up as being in Metaphase.

4 👍 | 👎 The daughter cells produced are genetically different from each other.

Answers on the next slide...

The Root Tip Experiment



Check your answers below



Laboratory Discussion on Cell Division

► Play audio

1



Dr. Miller explains that the meristem in the root tip is where growth is rapid.

2



Most cells are in Interphase, which accounts for about 90% of the cell cycle.

3



Chromosomes aligned in the middle is the hallmark of Metaphase.

4



Sarah mentions they are identical daughter cells.

When Mitosis Goes Wrong

The cell cycle is strictly controlled by genes. If a mutation occurs in these genes, the cell may start to divide uncontrollably.



Watch out

Uncontrolled cell division leads to the formation of a **tumour**, which is the basis of cancer.



Sequence the Cycle

Answers on the next slide...

Place the events of the cell cycle in the correct order, starting from the preparation phase.

Chromosomes line up and are pulled apart (Mitosis)

The cytoplasm and cell membrane divide (Cytokinesis)

DNA replication and growth (Interphase)

Two genetically identical daughter cells are formed

Sequence the Cycle



Place the events of the cell cycle in the correct order, starting from the preparation phase.

DNA replication and growth (Interphase)

1.

Chromosomes line up and are pulled apart (Mitosis)

2.

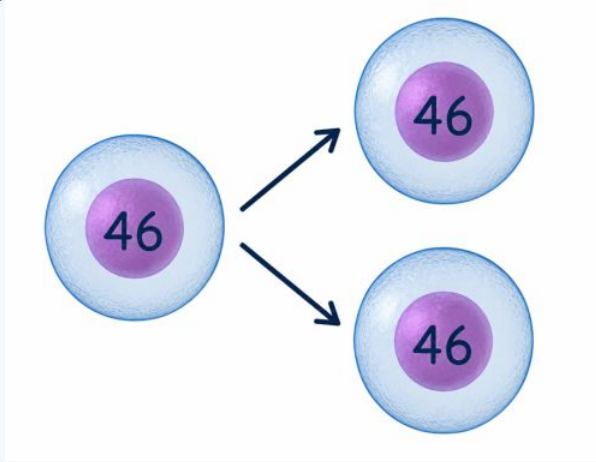
The cytoplasm and cell membrane divide (Cytokinesis)

3.

Two genetically identical daughter cells are formed

4.

Exam-Style Challenge



A parent cell has 46 chromosomes. After mitosis, how many chromosomes will each daughter cell have, and why is this specific number important for the organism?

Exam-Style Challenge



You might have said...

Each daughter cell will have 46 chromosomes.

It is important as daughter cells must be identical to the parent cell.

This ensures they perform the same function as the original.

It maintains the correct diploid number for the species.

Summary & Reflection

Key Takeaways

- **Mitosis** produces two genetically identical daughter cells.
- It is used for **growth, repair, and asexual reproduction**.
- DNA must be **replicated** during Interphase before division starts.
- Uncontrolled mitosis can result in **cancer**.

